

UI Design Implementation Exercises

Project Breakdown, Estimation & Execution Roadmap (React, RTK, TypeScript)

Estimated completion times depend on your current familiarity with **Redux Toolkit (RTK)** and **TypeScript**. The breakdown uses standard single-developer estimates, splitting hours between **UI/Styling** (HTML/CSS/ Layout) and **Logic/State Management** (React, RTK, TS typing).

Project 1: Interactive Kanban Task Board with Undo/Redo

Estimated: 12 – 16 Hours

Difficulty: Intermediate

PHASE	TIME	TASKS
UI & Styling	4 – 5 hrs	CSS Grid columns, card styling, drag visual feedback, dark mode switch
State & Logic	6 – 8 hrs	RTK state normalization (<code>createEntityAdapter</code>), undo/redo history stack
Types & Testing	2 – 3 hrs	Task/Board TS interfaces, keyboard shortcuts, edge-case debugging

Project 2: Real-Time Crypto/Stock Analytics Dashboard

Estimated: 14 – 18 Hours

Difficulty: Intermediate-Advanced

PHASE	TIME	TASKS
UI & Charts	5 – 6 hrs	Responsive CSS Grid, dynamic chart rendering (Chart.js / Recharts integration)
Data & Logic	7 – 9 hrs	RTK Query setup, automatic HTTP polling stream, watchlist state handlers
Types & Cleanup	2 – 3 hrs	API payload TS interfaces, flash animations on price changes

Project 3: E-Commerce Product Customizer & Checkout

Estimated: 10 – 14 Hours

Difficulty: Intermediate

PHASE	TIME	TASKS
UI & Transitions	4 – 5 hrs	Preview layer transitions, modal wizard layouts, responsive cart drawer
State & Selectors	4 – 6 hrs	RTK memoized selectors (<code>createSelector</code>), local storage syncing, step wizard state
Types & Validation	2 – 3 hrs	Form validation schemas, generic form step interfaces

Suggested Execution Path

Sprint 1 (Weekend Project) — Project 3

Start with Project 3 to build strong foundational patterns around forms, UI transitions, and RTK selectors.

Sprint 2 — Project 1

Build Project 1 to focus heavily on complex array operations, reordering logic, and custom Redux slice actions (undo/redo).

Sprint 3 — Project 2

Complete Project 2 to master data fetching lifecycle, caching, and real-time interface polling with RTK Query.

Recommended Tutorial

For a step-by-step walkthrough of building a task board UI with drag-and-drop interactions in React, refer to standard guides on building Kanban Boards in React for drag-and-drop state setup and component architecture.